

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

May 2012

CL303 Geotechnical Engineering-I

Date: 02.05.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain scope of geotechnical engineering. [03]

(b) The following results are obtained during a Standard Proctor Test. [04]

Moisture Content, %	5	10	20	24	30
Bulk Unit Weight, kN/m^3	17.2	19.4	21.2	21.6	21.7

Plot compaction curve and obtain OMC and MDD. Use $G=2.65$.

Q - 2 (a) Define MDD, OMC and Zero air void line with neat sketch. [04]

(b) In a consolidation test, the void ratio of the specimen which was 1.068 under effective pressure of 214 kN/m^2 changed to 0.994 when the pressure was increased to 430 kN/m^2 . Calculate the compression index. [05]

(c) Explain how to determine the pre-consolidation pressure? [05]

OR

Q - 2 (a) Distinguish between standard proctor test and modified proctor test. [04]

(b) Define compression index, swelling index, time factor and degree of consolidation. [05]

(c) In a laboratory consolidation test on a 20 mm thick sample of saturated clay taken from a site, 50% consolidation was reached in 10 minutes with double drainage. Estimate the time required for the clay layer of 5m thickness at the site for 50% consolidation, if there is drainage only towards the top. [05]

Q - 3 (a) What is primary and secondary consolidation? [02]

(b) List the tests for determination of shear strength of soil. Explain direct shear test. [06]

(c) A soil sample has total normal stress = 200 kN/m^2 , pore water pressure = 80 kN/m^2 , $C=16 \text{ kN/m}^2$ and $\phi = 30^\circ$. Determine effective shear strength. [06]

OR

Q - 3 (a) A layer of clay 5m thick lies under a newly constructed building. The effective pressure due to overlying strata on the clay layer is 3.0 kg/cm^2 and the new construction increases the overburden by 1.2 kg/cm^2 . If the compression index of clay is 0.45, compute the settlement assuming the natural water content of the clay layer to be 43% and the specific gravity of its soil grains as 2.7. [02]

(b) Draw Triaxial test soil specimen; give its dimensions, show deviator stress and Confining pressure on it. [06]

(c) Explain unconfined compression test for shear strength of soil [06]

SECTION - II

Q - 4 (a) The soil is having Liquid Limit = 40%, Plastic Limit = 10%. [04]

% passing 4.75mm sieve = 70%;

% passing 75 μ sieve = 35%.

Classify the soil based on USC classification system and give its symbolic representation and typical name. Also plot the plasticity chart using appropriate scale.

- (b) What is soil structure? Explain types of structure for clayey soils with figures. [03]
- Q - 5 (a) Define the following: (Any Three) [03]
- (1) Alluvial soils (2) Colluvial soils (3) Aeolian soils (4) Humus (5) Loam
- (b) (1) Define the terms Uniformity coefficient and Coefficient of curvature. [02]
- (2) Explain uses of particle size distribution curve. [03]
- (c) An undisturbed sample of soil has a volume of 100 cm^3 and mass of 190 gram. On oven drying for 24 hours; the mass is reduced to 160 gram. The specific gravity of the solid is 2.68. Determine the Bulk Density, Dry Density, Water content, Void ratio, Porosity, Degree of saturation. [06]

OR

- Q - 5 (a) Explain soil formation in geological cycle. [03]
- (b) (1) Distinguish between coarse grained soils and fine grained soils. [02]
- (2) An oven-dried soil having a mass of 200 gram is placed in a pycnometer which is then completely filled with water. The total mass of the pycnometer with water and soil inside is 1605 gram. The pycnometer filled water alone has a mass of 1480 gram. Calculate the specific gravity of the soil. [03]
- (c) A sample of sand above water table was found to have a natural moisture content of 15% and a unit weight of 18.84 kN/m^3 . Laboratory tests on a dried sample indicated values of 0.5 and 0.85 for minimum and maximum void ratios respectively for densest and loosest states. Calculate the degree of saturation and relative density. Assume $G=2.65$. [06]
- Q - 6 (a) (1) Derive an expression for dry unit weight in terms of bulk unit weight and water content. [03]
- (2) Give the nomenclature of soil particles as per Indian Standard. [02]
- (b) In a falling head permeameter, the initial head ($t=0$) is 40 cm. The head drops by 5 cm in 10 minutes. Calculate the time required to run the test for the final head to be at 20 cm. If the sample used is 6cm is height and 50 cm^2 in cross-sectional area. Calculate the coefficient of permeability, taking area of stand pipe $=0.5 \text{ cm}^2$. [09]

OR

- Q - 6 (a) (1) Derive an expression for bulk unit weight in terms of its void ratio, degree of saturation, specific gravity of solids and unit weight of water. [03]
- (2) One kg of soil was sieved through a set of 8 sieves with size 4.75mm, 2.0 mm, 600μ , 425μ , 300μ , 212μ , 150μ , and 75μ . The mass of soil retained on these sieves was found to be 50, 78, 90, 150, 160, 145, 148 and 179 gm respectively. Determine the percentage finer than the corresponding sizes. [02]
- (b) (1) Enlist factors affecting the permeability and describe any two. [03]
- (2) A horizontal stratified soil deposit consists of three uniform layers of thickness 6, 4, and 12 m respectively. The permeabilities of these layers are $k_1= 8 \times 10^{-4} \text{ cm/sec}$, $k_2= 52 \times 10^{-4} \text{ cm/sec}$, $k_3=6 \times 10^{-4} \text{ cm/sec}$. Find the effective average permeability of the deposit in the horizontal and vertical direction. [06]
